

## TABLE OF CONTENTS

|                                                                                       |           |
|---------------------------------------------------------------------------------------|-----------|
| <b>TABLE OF CONTENTS .....</b>                                                        | <b>1</b>  |
| <b>1- INTRODUCTION .....</b>                                                          | <b>2</b>  |
| 1-1 Getting Acquainted, The Computer From The Front.....                              | 2         |
| 1-2 Getting Acquainted, The Computer From The Front, Front Bezel Removed.....         | 3         |
| <b>2- OCTANE 2 MR FRU LEVELS .....</b>                                                | <b>5</b>  |
| <b>3- WARNINGS AND CAUTIONS BEFORE YOU BEGIN .....</b>                                | <b>6</b>  |
| 3-1 System Identification Module.....                                                 | 6         |
| 3-2 Static Electricity Sensitive Device.....                                          | 6         |
| 3-3 SCSI Cabling and System Disk (Disk unit 1) requirements .....                     | 6         |
| <b>4- REPLACING THE OCTANE 2 COMPUTER.....</b>                                        | <b>8</b>  |
| <b>5- CORRECTING VERTICAL ROLL DUE TO FIX LINE RATE MONITOR INCOMPATABILITY .....</b> | <b>9</b>  |
| 5-1 Adjusting The Default Frequency Rate of the Octane II Video Board.....            | 9         |
| A-1 Signa Power up Delay.....                                                         | 11        |
| A-2 [hinv] Command .....                                                              | 11        |
| <b>REVISION HISTORY .....</b>                                                         | <b>12</b> |

## 1- INTRODUCTION

The standard configuration Octane2 as used by GEMS MR consists of single 360 MHz R12000A processor. V10 single head or V2 dual head image processor.

### 1-1 Getting Acquainted, The Computer From The Front

Illustration 1-1 shows the front of the Octane 2 box.

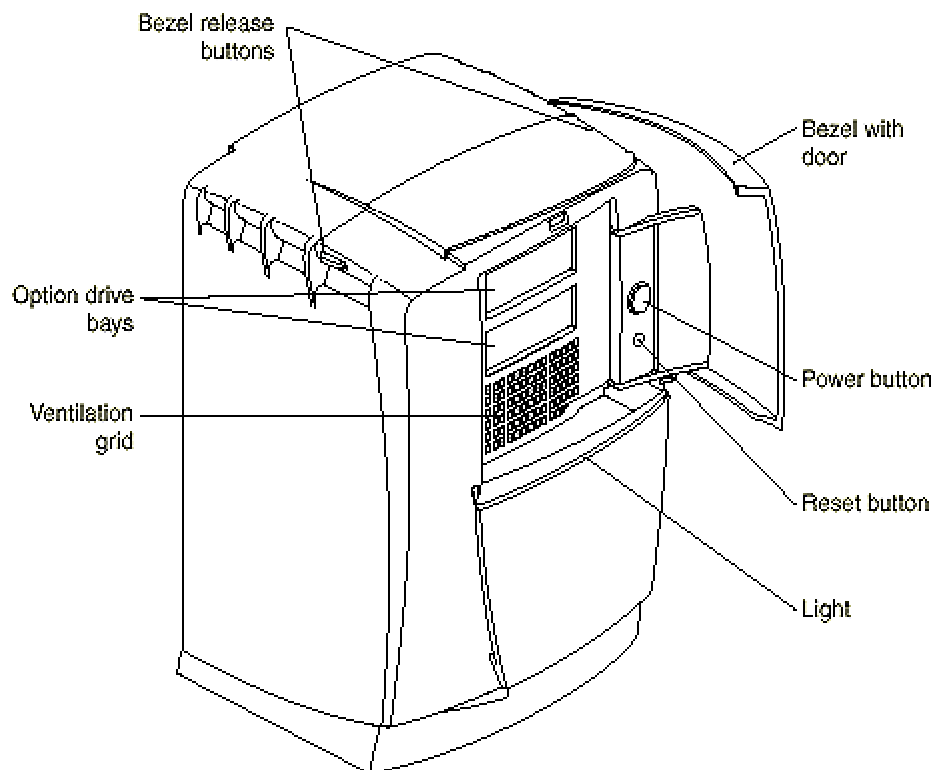


Illustration 1-1  
Octane II, A Front View

## 1-2 Getting Acquainted, The Computer From The Front, Front Bezel Removed

Illustration 1-2 Shows the Computer from the front. The front bezel removed to show drive bays and light bar. The Illustration shows one disk drive installed. Your system may have two.

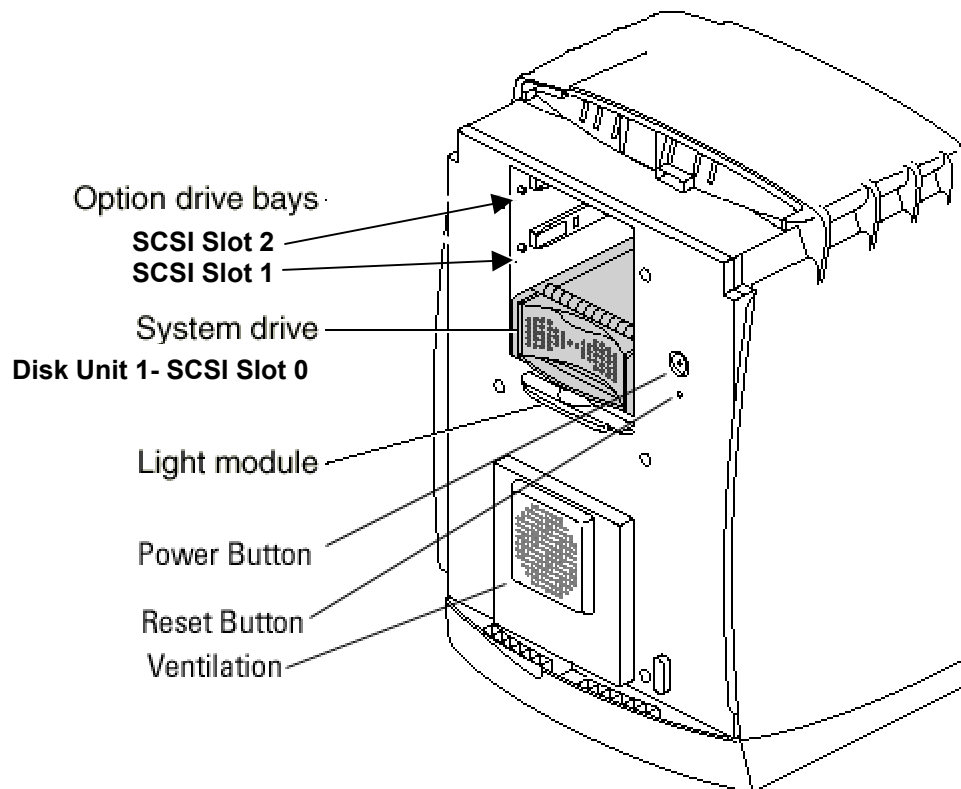


ILLUSTRATION 1-2  
COMPUTER FRONT, BEZEL REMOVED

### Note:

The system software and the Software Load process requires that Disk unit 1 (System Disk) be in SCSI Slot zero (0). The Signa software is hard coded to look for the System Disk in Slot zero (0). If the disk drive is not in this location, the software load process will fail as well as any attempt to re-load software.

### Note:

Replacement of the Octane 2 requires disconnection of the SCSI cables at the rear of the unit. The SCSI cables are not marked. Ensure that the correct SCSI cable for Disk drive 1 is reconnected to slot zero (0) of the Octane 2. The System Software will fail to load (Unable to locate System Disk error will appear) if the wrong SCSI cable is connected to Slot zero (0). The System Disk must be connected to Slot zero (0).

1-3 Getting Acquainted, The Computer From The Back  
Illustration 1-3 shows the rear of the Octane II box.

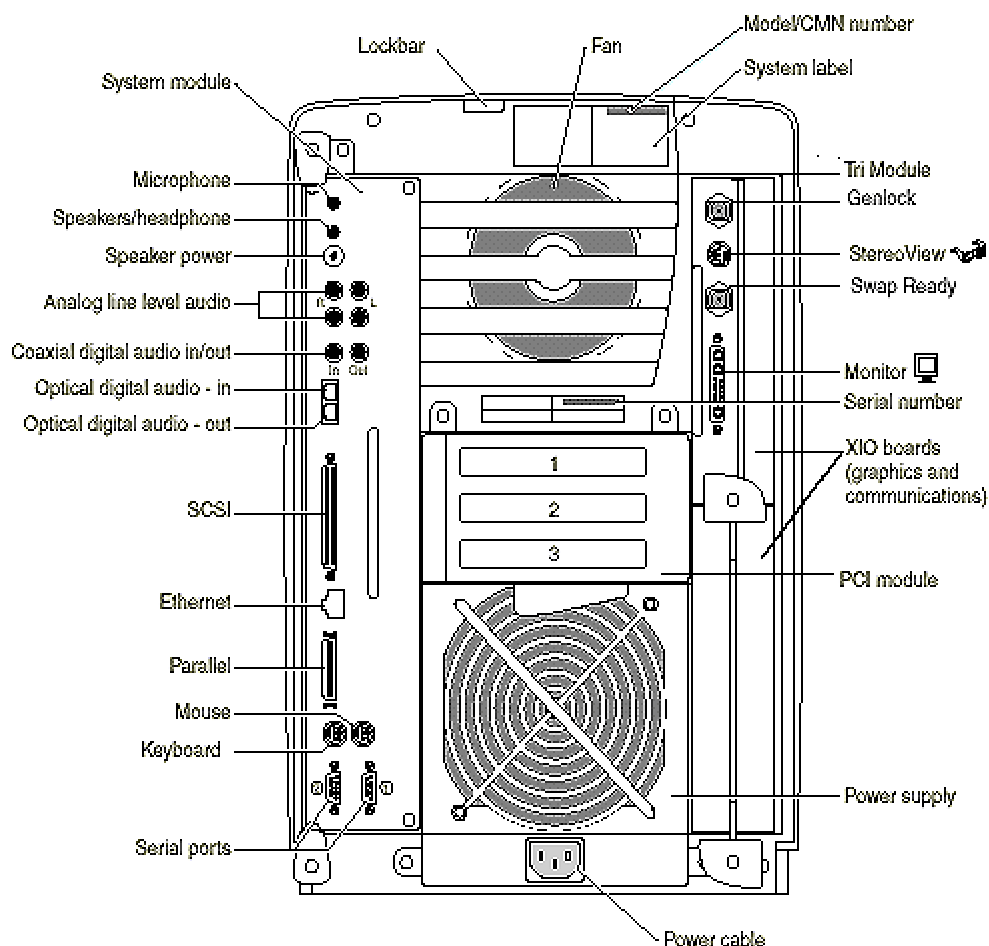


ILLUSTRATION 1-3  
OCTANE 2 REAR VIEW

## 2- OCTANE 2 MR FRU LEVELS

Table 2-1 and 2-2 give the breakdown of FRU levels for the Octane 2.

TABLE 2-1  
OCTANE 2 USING V10 GRAPHICS FRU LEVELS, NORMAL MR CONFIGURATION

| Part #            | Description (Normal MR Configuration)                                                                                                                                                                                                                                                                                                                                                                                                              | Comment        |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| 2200169-3         | OCTANE II, MR SINGLE PROC UNIT, V10<br>This FRU contains the entire Octane2 Box with the following parts Installed:<br><ul style="list-style-type: none"> <li>• Octane/Octane2 Power Supply Module (Cherokee)</li> <li>• Octane/Octane2 Front Light Bar</li> <li>• IP30 motherboard (No Memory Installed)</li> <li>• Single R12K-360MHZ w/2MB L2 Processor Module</li> <li>• Octane2 V10 Graphics Module</li> <li>• Octane2 Front Plane</li> </ul> | T3 FRU         |
| 2199806-8         | 0.5GB memory sticks                                                                                                                                                                                                                                                                                                                                                                                                                                | Consumable FRU |
| 2169940-7(2 each) | 9GB Disk Drive                                                                                                                                                                                                                                                                                                                                                                                                                                     | Consumable FRU |
| 2300173-2         | MR PCI CARD CAGE ASSEMBLY (collector) which contains:<br><ul style="list-style-type: none"> <li>• Option PCI Card cage</li> <li>• Option PCI SCSI card.</li> <li>• PCI Adapter card.</li> <li>• Encore Fibre Optic Strain</li> </ul>                                                                                                                                                                                                               | T3 FRU         |
| 2169940-26        | Octane System ID Button (NIC module)                                                                                                                                                                                                                                                                                                                                                                                                               | Consumable FRU |

TABLE 2-2  
OCTANE 2 USING V12 GRAPHICS FRU LEVELS

(This information is for reference only. MR configuration does not support V12 Graphic Card)

| Part #     | Description                                                                                                                                                                                                                                                                                                                                                                                                                                         | Comment        |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| 2300169-4  | OCTANE 2, MR SINGLE PROC UNIT, V12<br>This FRU contains the entire Octane2 Box with the following parts Installed:<br><ul style="list-style-type: none"> <li>• Octane/Octane 2 Power Supply Module (Cherokee)</li> <li>• Octane/Octane 2 Front Light Bar</li> <li>• IP30 motherboard (No Memory Installed)</li> <li>• Single R12K-360MHZ w/2MB L2 Processor Module</li> <li>• Octane2 V12 Graphics Module</li> <li>• Octane2 Front Plane</li> </ul> | T3 FRU         |
| 2199806-8  | 0.5GB memory sticks                                                                                                                                                                                                                                                                                                                                                                                                                                 | Consumable FRU |
| 2169940-72 | 9GB Disk Drive                                                                                                                                                                                                                                                                                                                                                                                                                                      | Consumable FRU |
| 2300173-2  | MR PCI CARD CAGE ASSEMBLY (collector) which contains:<br><ul style="list-style-type: none"> <li>• Option PCI Card cage</li> <li>• Option PCI SCSI Card</li> <li>• PCI Adapter Card</li> <li>• Encore Fiber Optic Strain</li> </ul>                                                                                                                                                                                                                  | T3 FRU         |
| 2169940-26 | Octane System ID Button (NIC module)                                                                                                                                                                                                                                                                                                                                                                                                                | Consumable FRU |

### 3- WARNINGS AND CAUTIONS BEFORE YOU BEGIN

#### 3-1 System Identification Module

#### **WARNING!**

IF THE OCTANE 2 OR JUST THE FRONT PLANE MODULE IS CHANGED OUT, THE SYSTEM IDENTIFICATION MODULE MUST BE REMOVED FROM THE OLD COMPUTERS FRONT PLANE MODULE BEING REPLACED AND PUT INTO THE NEW COMPUTER BEING INSTALLED. IF THE SYSTEM IDENTIFICATION MODULE FROM THE OLD COMPUTER BEING REPLACED IS NOT INSTALLED INTO THE NEW COMPUTER, CUSTOMER OPTION KEYS WILL NO LONGER FUNCTION AND WILL NOT RELOAD. NEW OPTION KEY MOD'S WILL HAVE TO BE ORDERED FOR THE SITE IN ORDER TO REINSTALL CUSTOMER PAID OPTIONS.

#### 3-2 Static Electricity Sensitive Device

#### **CAUTION**

The components of this computer are extremely sensitive to static electricity. You must wear a grounding wrist strap while working with any module inside this computer. See Illustration 3-1. Also use of a grounding mat if parts must be laid down flat to work on.

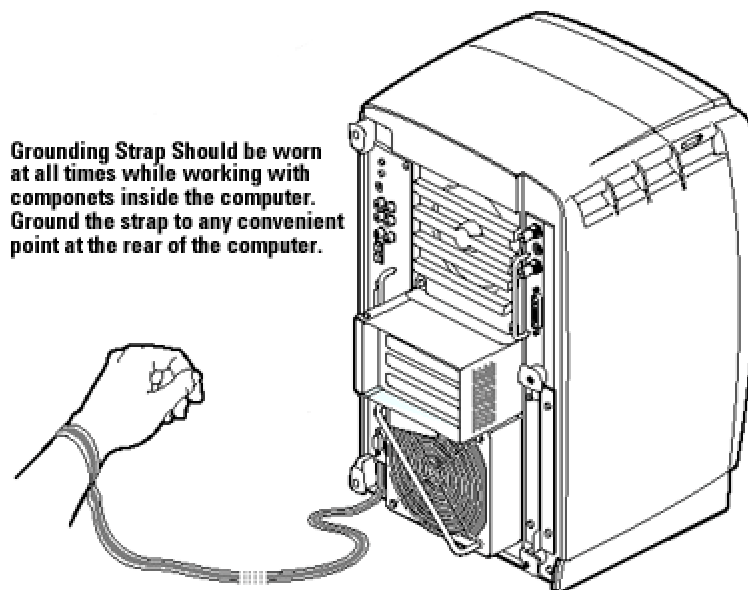


ILLUSTRATION 3-1  
WEARING A GROUNDING STRAP

#### 3-3 SCSI Cabling and System Disk (Disk unit 1) requirements



**WARNING!**

If it is possible, it is recommended that prior to removal of any of the SCSI cables or Disk drives in the Octane 2 computer, that the "hinv" (Hardware inventory) command be run in a c-shell and all the information reported for the disk drive locations noted.

**How to run hinv**

1. If possible power up the computer
2. Open up a C-shell
3. type: hinv <enter>
4. Note the information found related to the disk drives. (See appendix A)

**Note:**

Replacement of the Octane 2 requires disconnection of the SCSI cables at the rear of the unit. The SCSI cables are not marked. Ensure that the correct SCSI cable for Disk drive 1 is reconnected to slot zero (0) of the Octane 2. The System Software will fail to load (Unable to locate System Disk error will appear) if the wrong SCSI cable is connected to Slot zero (0).

#### 4- REPLACING THE OCTANE 2 COMPUTER

Octane2 FRU Levels are defined in [Section 2](#). Refer to illustration 4-1 for the basic steps involved in changing out the Octane 2 Computer. Refer to the attached vendors manual for directions on individual component removal and replacements.

##### Note

Signa systems with a fixed line rate of 72Hz, (ALL older CRT's, Models MDW 321 and MDW321+) will not sync properly with the Octane II Video board. The monitor will vertically "roll" continuously. Refer to **Section 5** for instructions on how to correct this problem.

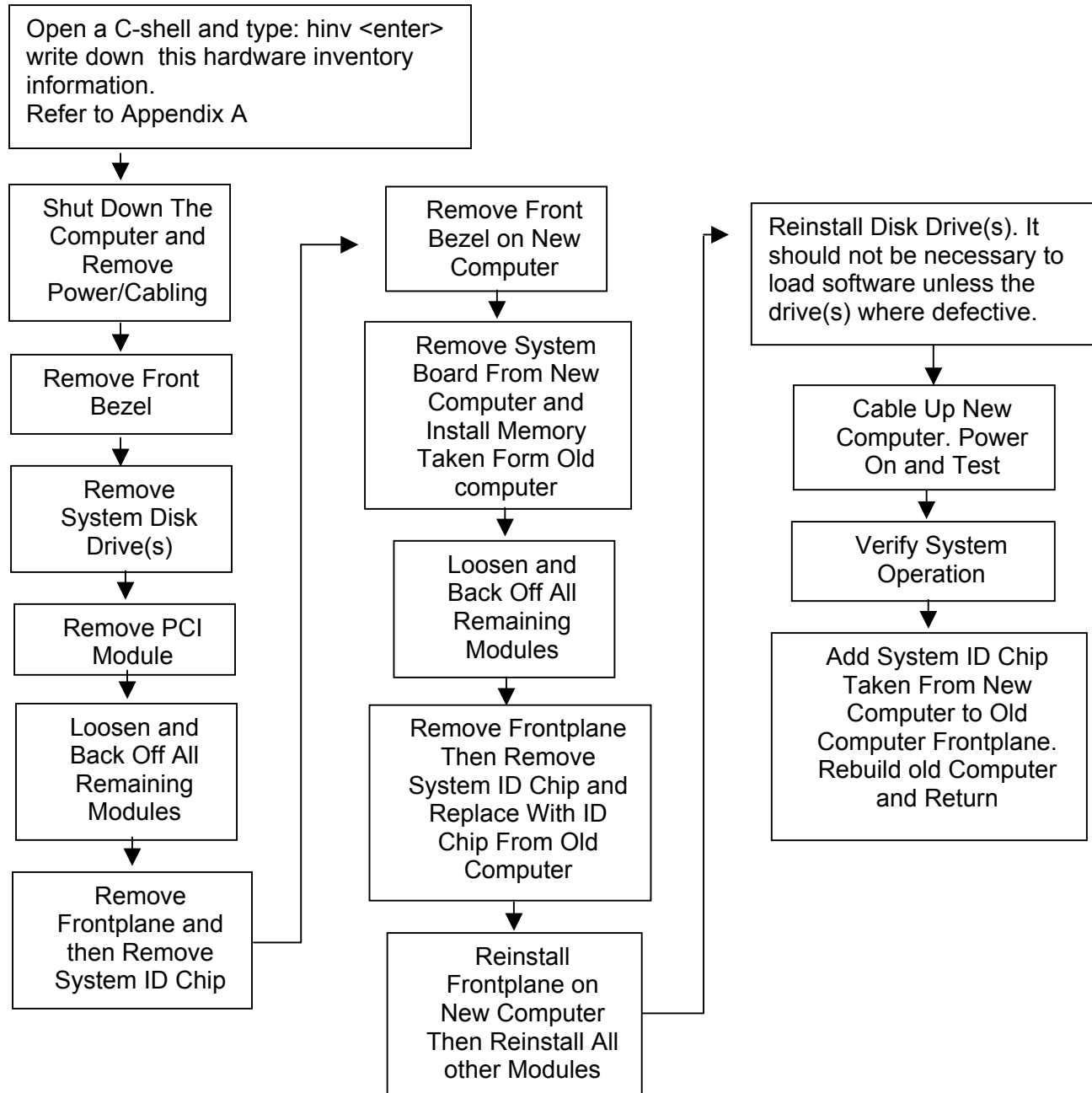


ILLUSTRATION 4-1  
REPLACING THE OCTANE2 FRU COMPUTER



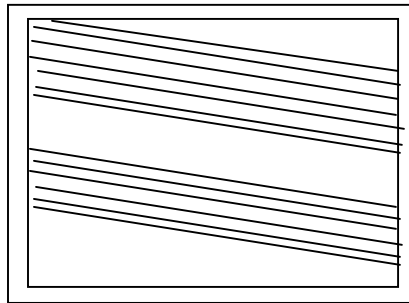
For further detail on the SGI Octane2 refer to the Vendors Manual

## 5- CORRECTING VERTICAL ROLL DUE TO FIX LINE RATE MONITOR INCOMPATABILITY

The Line Rate Frequency of the video board in the Octane II is set at factory default of 60Hz. This line rate is NOT compatible with some older CRT Fixed Line Rate monitors on earlier LX systems. This section explains how to change the Octane II video boards' default line rate to 72Hz.

### 5-1 Adjusting The Default Frequency Rate of the Octane II Video Board

1. Make sure the CRT monitor is connected to the Octane II computer and power is applied. The monitor will be rolling vertically at this time.



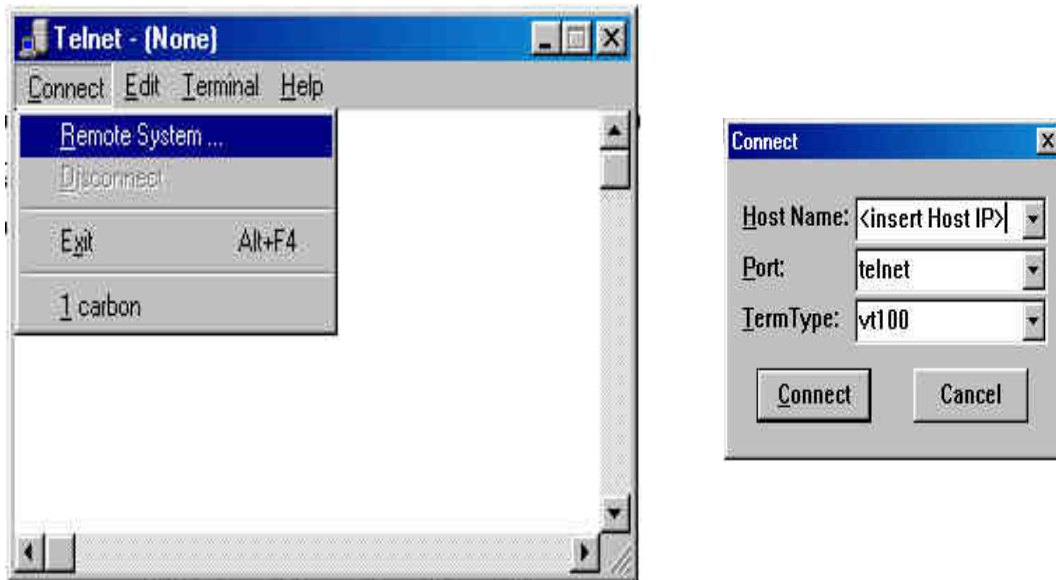
MONITOR RASTER ROLLING VERTICALLY OUT OF SYNC  
ILLUSTRATION 5-1

2. Switch the keyboard over to the PC.
3. In Microsoft Windows Select:
  - a. **Start**
  - b. **Run**



STARTING A TELNET SESSION  
ILLUSTRATION 5-2

4. Type **telnet** <enter> (Start a Telnet Session)
5. Click on **Connect** drop down menu.
6. Select **Remote System**



INSERTING THE HOST IP ADDRESS  
ILLUSTRATION 1-5

7. Type: **The Host IP (Internet address) of your Host (SGI) computer.** (example: 192.1.1.1)
8. Select **Connect** button.
9. login: **root** <enter> password **operator** <enter>
10. Type: **cd /usr/gfx** <enter>
11. Type: **setmon -x 1280x1024\_72** <enter>
12. Push the power off button on the host (SGI) computer. (The Light bar will flash red/white for approximately 1 minute and then the Host (SGI) computer will shut off.
13. Push the power on button on the Host (SGI) computer to restart the system.
14. Reboot Signa
15. This will permanently change the Video Line Rate of the Video board to 72Hz.

**Note**

Should the software be reloaded, or the IP30 (Mainboard) or Octane II be replaced, this procedure will need to be re-performed.

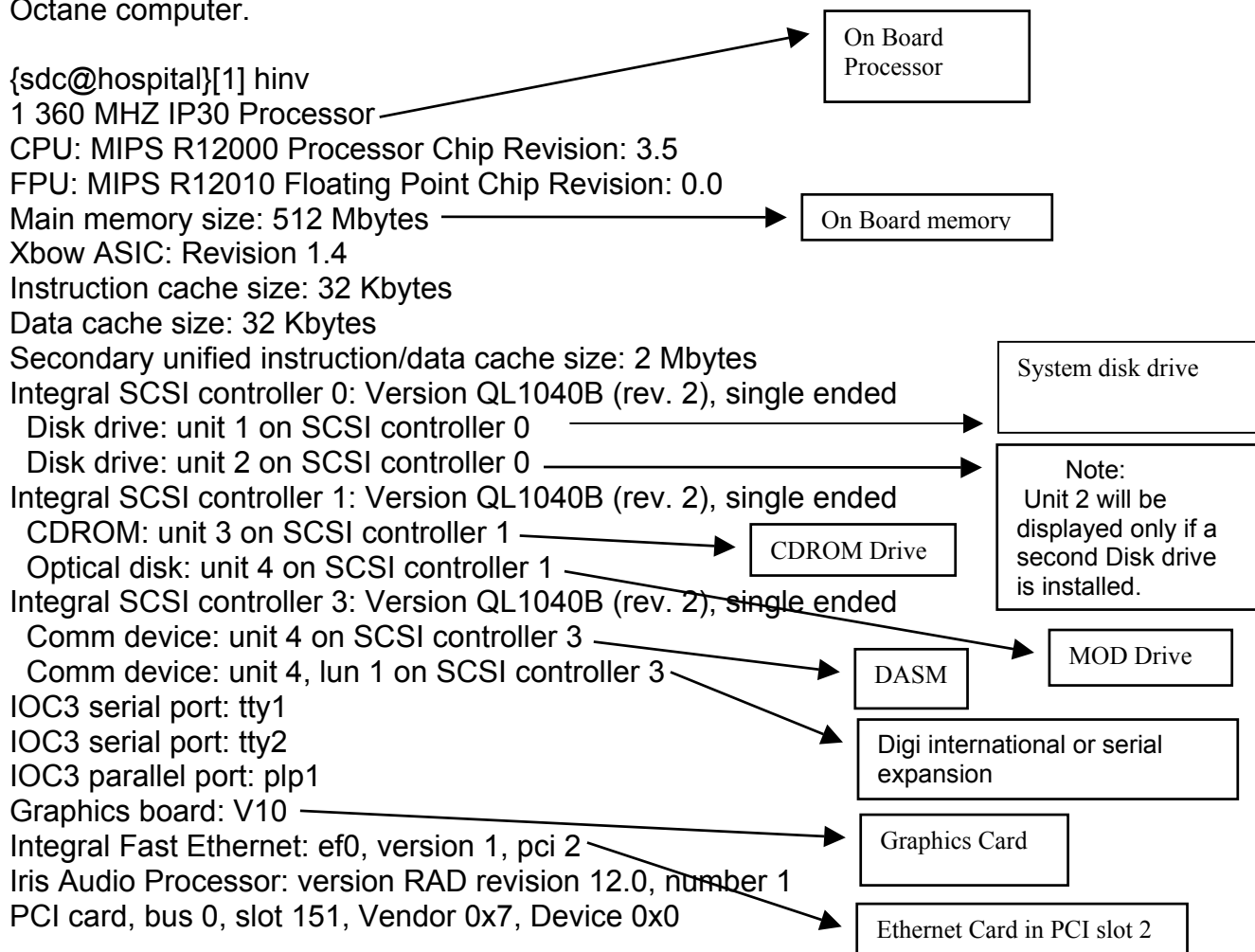
## A- APPENDIX - hinv REPORT OUTPUT

### A-1 Signa Power up Delay

When Signa is first powered on there is a considerable period of time that nothing is seen on the monitor; up to 4 minutes depending on the computer type. The system is checking configurations and starting software processes at this time. If you are concerned that the system is hung, push the **<Esc>** button on the top left side of the keyboard. The system will then post to the monitor all its boot up activities.

### A-2 [hinv] Command

The **[hinv]** command can be used to check the configuration of the operator workspace. To use the command, open a command window and at the prompt type: **hinv <Enter>**. The following is an example of an **[hinv]** output of a common configured Signa system using Octane computer.



## REVISION HISTORY

| REV | DATE        | AUTHOR        | PRIMARY REASONS FOR CHANGE                                                        |
|-----|-------------|---------------|-----------------------------------------------------------------------------------|
| A   | 10/29/2001  | Hawthorne     | Initial Version.                                                                  |
| B   | 10/29/01    | D. Hofstetter | Added references to SCSI cable connections and "hinv" command output. Appendix A. |
| C   | Nov 6, 2001 | D. Hofstetter | Added Section 5. Adjustment of Octane II Video Boards' Line rate.                 |
|     |             |               |                                                                                   |
|     |             |               |                                                                                   |